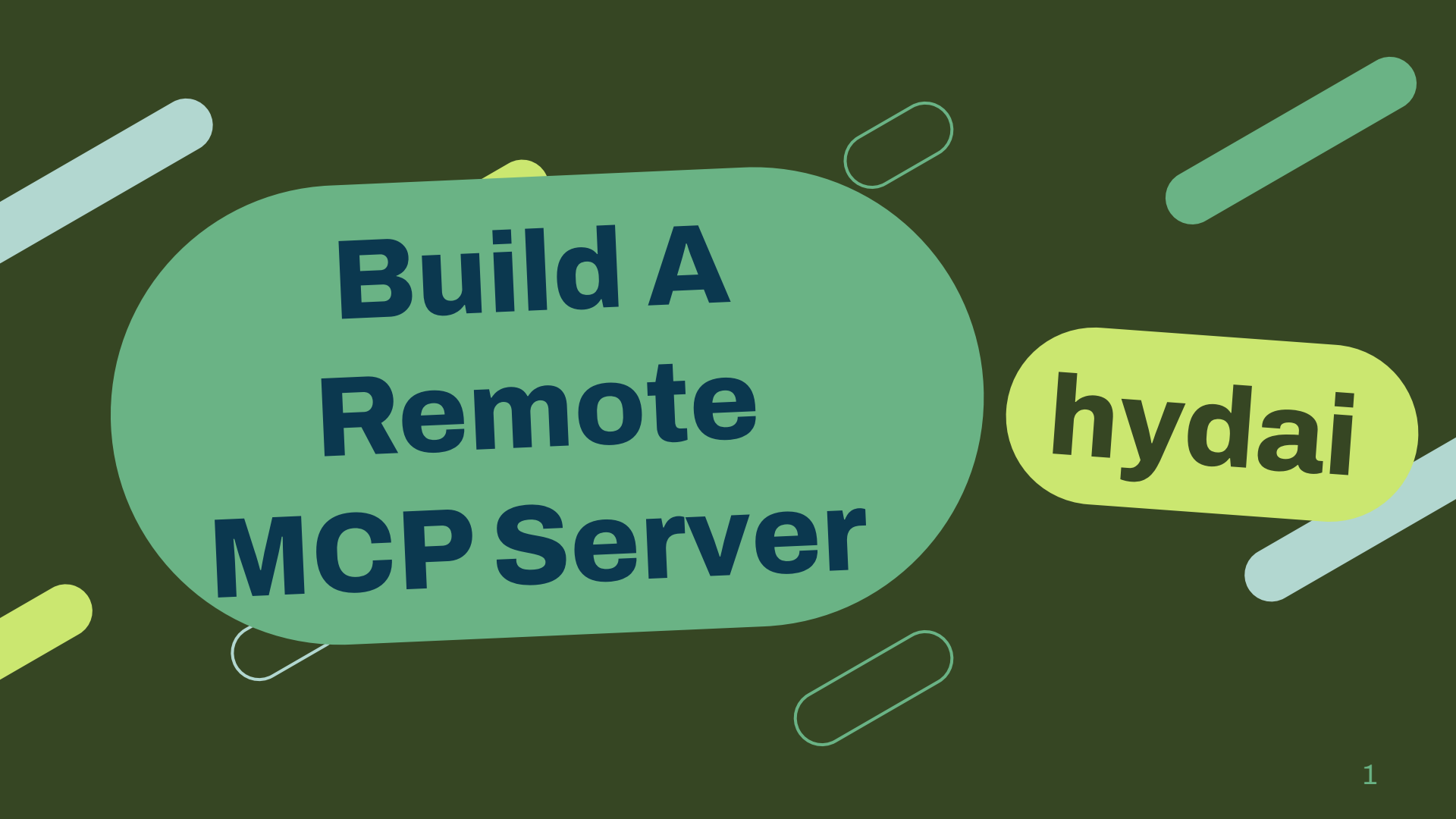




Build A Remote MCP Server

hyundai

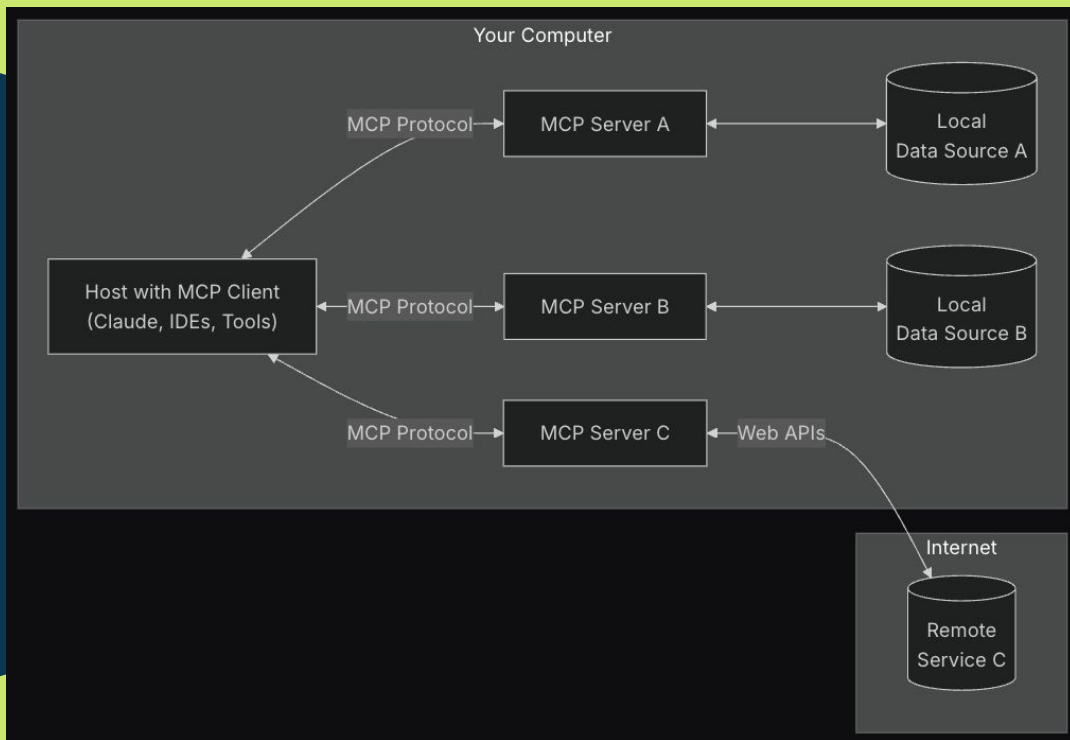


hydai

WasmEdge Maintainer
CNCF Ambassador
EVM/Solidity Contributor

<https://url.hyd.ai/remotemcp>

Model Context Protocol



MCP Hosts

The application that users interact with.

Cline

Claude Desktop

IDE

AI Tools

MCP Clients

Programs maintain 1:1 connections with
servers

MCP Servers

Lightweight programs that each expose specific capabilities through the standardized Model Context Protocol

Local Data Sources

Your computer's files, databases, and services that MCP servers can securely access

DB

FileSystem

Unity

Browser

Remote Services

External systems available over the internet
(e.g., through APIs) that MCP servers can
connect to

GSuite

Notion

Weather

GitHub

MCP Servers

Capabilities

Resources: File-like data

Tools: Callable functions

Prompts: Pre-written templates

Resources

Tools

Prompts

How to interact

STDIO: Local

SSE: Local/Remote

(3/26 updated) Streamable HTTP:
Local/Remote

STDIO

SSE

HTTP

STDIO

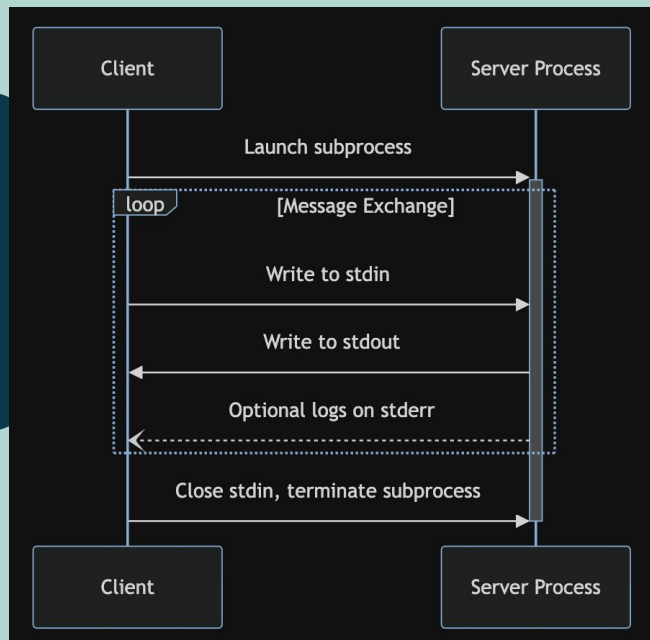
`npx @x/y args`

`uvx x args`

`wasmedge x.wasm args`

`docker run`

STDIO



SSE Server

`npm start # TS`

`./server.bin # Rust`

`wasmedge sse.wasm args`

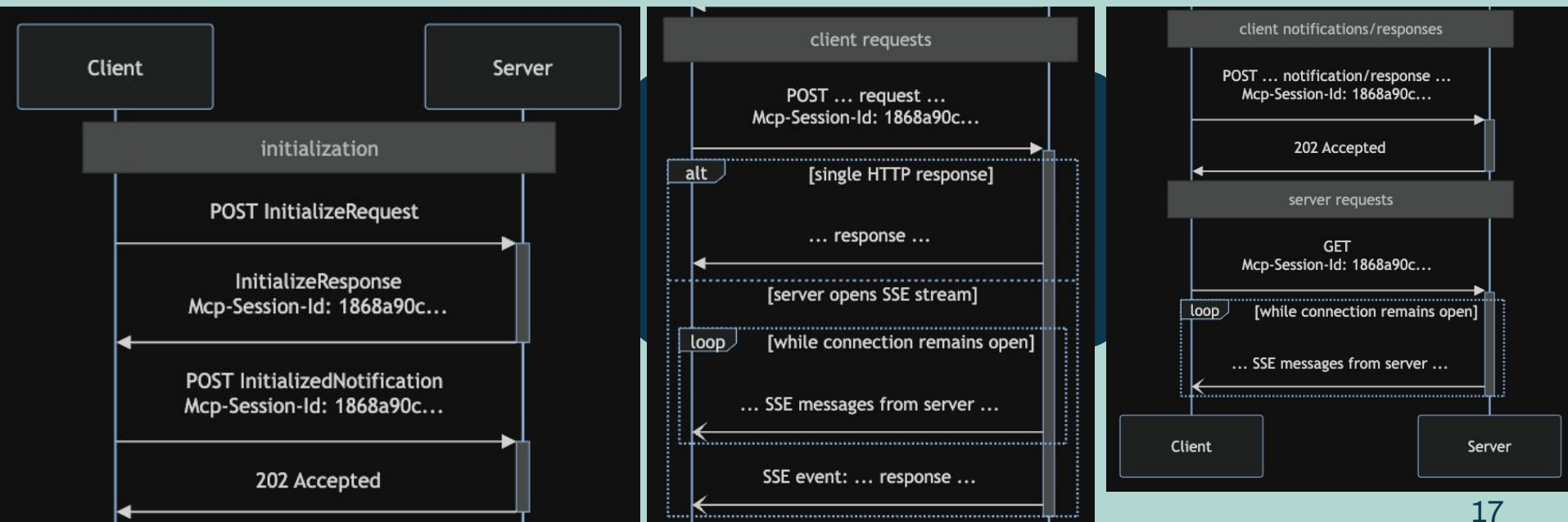
SSE Client

`npx mcp-remote URL`

mcp-remote

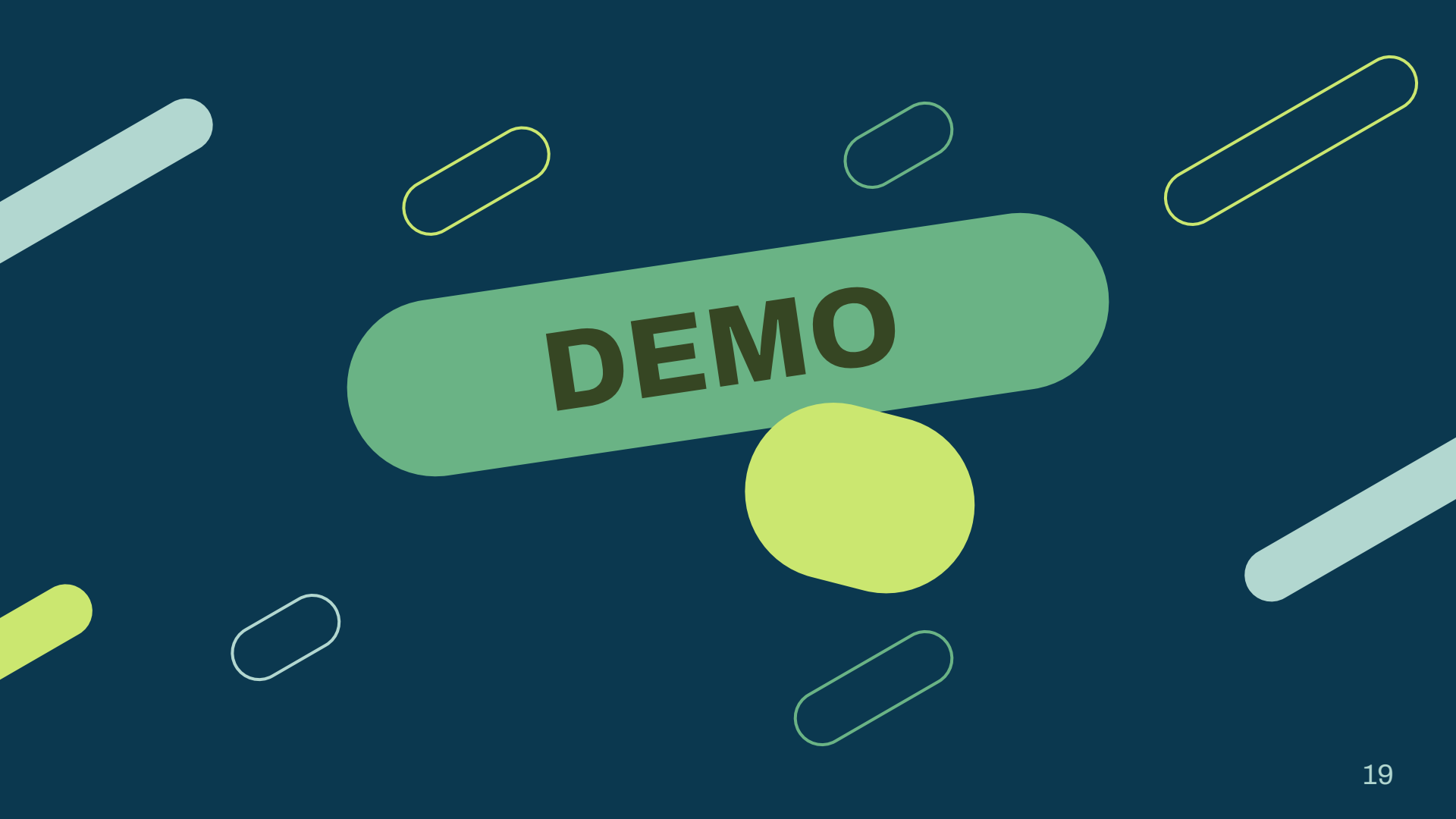
<https://www.npmjs.com/package/mcp-remote>

SSE



HTTP

New spec from 3/26.
Everything is a work in progress.

The image features a dark blue background with various geometric shapes. In the center, a light green rounded rectangle contains the word "DEMO" in bold, dark brown capital letters. Below this rectangle is a bright yellow circle. Scattered around these central elements are several light blue rounded rectangles and yellow-outlined rounded rectangles, all tilted at different angles.

DEMO

STDIO

MCP Rust SDK -> Wasm

```
git clone -b demo https://github.com/hydai/mcp-rust-sdk.git
cd mcp-rust-sdk/examples/servers
cargo build -r --example wasi_std_io --target wasm32-wasip1
wasmedge --dir .:. \
../..target/wasm32-wasip1/release/examples/wasi_std_io.wasm
```

Clients

```
{
  "mcpServers": {
    "wasm-demo": {
      "command": "wasmedge",
      "args": [
        "--dir=.:<path/to/logs>", # For logs
        "<path/to/wasm>"
      ]
    }
  }
}
```

SSE

MCP Rust SDK -> Native Binary

```
git clone https://github.com/modelcontextprotocol/rust-sdk.git
cd rust-sdk/examples/servers
cargo build -r --example axum
../../target/release/examples/axum
```

Clients

```
{  
  "mcpServers": {  
    "native-demo": {  
      "command": "npx",  
      "args": [  
        "mcp-remote",  
        "http://localhost:8000/sse"  
      ]  
    }  
  }  
}
```

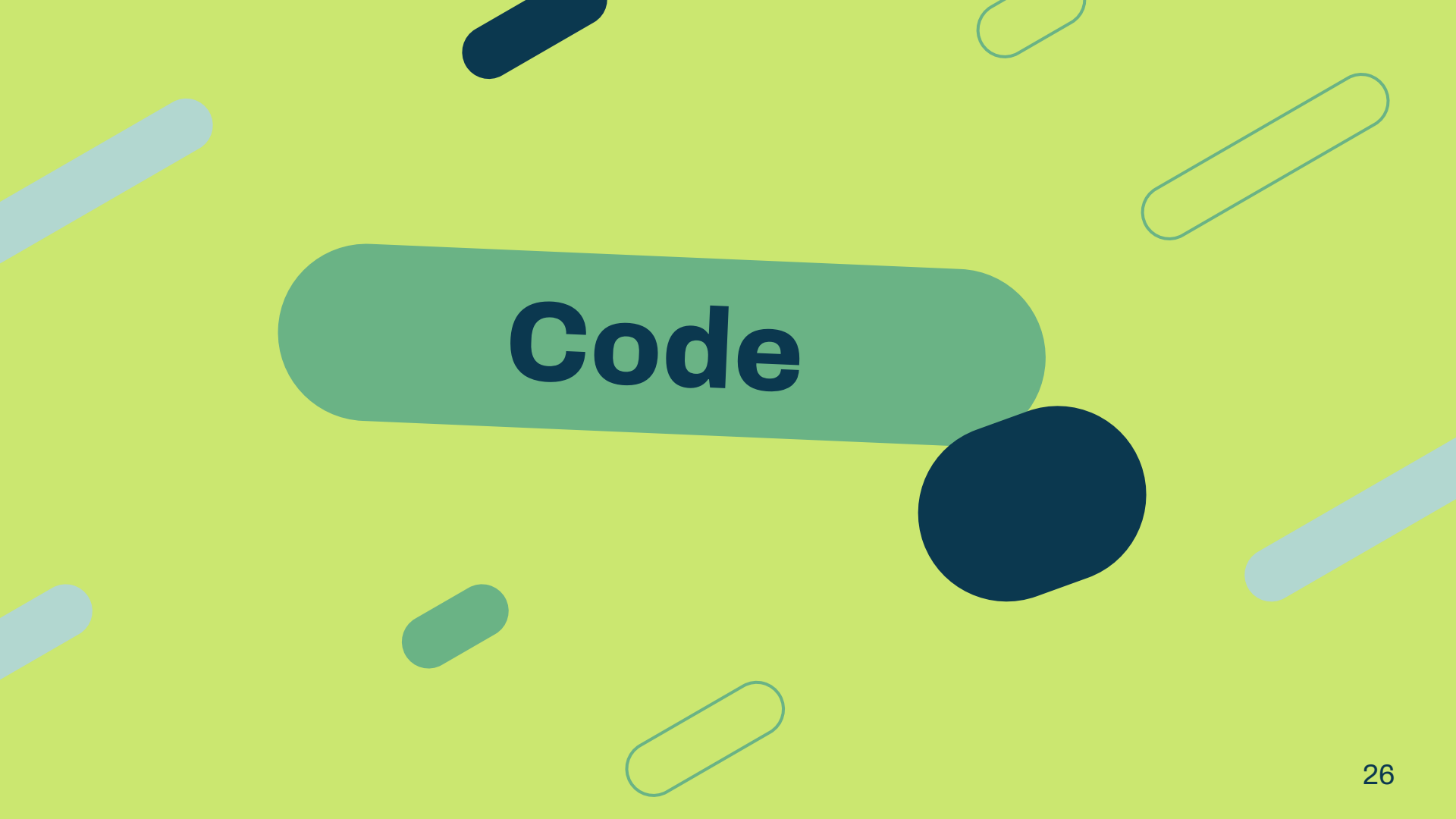
SSE

CloudFlare Remote MCP SDK

```
git clone git@github.com:cloudflare/ai.git  
cd ai  
npm install  
npx nx dev remote-mcp-server
```


Clients

```
{  
  "mcpServers": {  
    "cf-demo": {  
      "command": "npx",  
      "args": [  
        "mcp-remote",  
        "http://localhost:8787/sse"  
      ]  
    }  
  }  
}
```

The background is a solid light green color. It is decorated with several abstract geometric shapes: a large dark blue circle on the right side, a large teal rounded rectangle in the center containing the word 'Code', and several smaller rounded rectangles and circles in shades of blue and teal scattered around. Some of these shapes are solid, while others are outlined.

Code

THANKS

Q&A